Zeguan Wu

University of Pittsburgh
210 S. Bouquet Street
Pittsburgh, PA 15260
Email: zew79@pitt.edu
Last updated: April, 2026

Research Interests

Quantum computing and mathematical optimization; AI-assisted mathematical discovery and formal verification; agentic workflows for quantum algorithm design and validation.

Education

Lehigh University 2025
Ph.D., Industrial and Systems Engineering Bethlehem, PA

Advisor: Tamás Terlaky and Xiu Yang.
Committee: Arielle Carr, Carleton Coffrin, and Michael Saunders.

Columbia University 2019
M.Sc., Operations Research New York, NY

Nanjing University 2018
B.Sc., Material Physics Nanjing, China

Work Experience

University of Pittsburgh 2025–Present
NASA Postdoc Fellow Pittsburgh, PA

Pacific Northwest National Laboratory 2025–2025
Quantum Computing Internship Richland, WA

Los Alamos National Laboratory 2023–2025
Graduate Research Assistant Los Alamos, NM

Preprints and Papers Under Review

1. "Infeasibility Aware Large Language Models for Combinatorial Optimization", Yakun Wang, Min Chen, Zeguan Wu*, Junyu Liu, Sitao Zhang, and Zhenwen Shao. *arXiv:2604.01455*, 2026.

2. "Artificial Entanglement in the Fine-Tuning of Large Language Models", Min Chen, Zihan Wang, Canyu Chen, Zeguan Wu*, Manling Li, and Junyu Liu. *arXiv:2601.06788*, 2026.

3. "Provably efficient quantum algorithms for solving nonlinear differential equations using multiple bosonic modes coupled with qubits", Yu Gan, Hirad Alipanah, Jinglei Cheng, Zeguan Wu*, Guangyi Li, Juan José Mendoza-Arenas, Peyman Givi, Mujeeb R Malik, Brian J McDermott, and Junyu Liu.

4. "Stab-QRAM: an all-Clifford quantum random access memory for special data", Guangyi Li, Yu Gan, Zeguan Wu*, Xueyue Zhang, Zheshen Zhang, and Junyu Liu.

5. "An efficient quantum algorithm for linear system problem in tensor format", Zeguan Wu*, Sidhant Misra, Tamás Terlaky, Xiu Yang, and Marc Vuffray.

Publication

1. "A preconditioned inexact infeasible quantum interior point method for linear optimization", Zeguan Wu*, Xiu Yang, and Tamás Terlaky.
Computational Optimization and Applications, 93(3):1225–1257, 2026.

2. "A quantum dual logarithmic barrier method for linear optimization", Zeguan Wu*, Pouya Sampourmahani, Mohammadhossein Mohammadisiahroudi, and Tamás Terlaky.
INFORMS Journal on Optimization.

3. "An inexact feasible interior point method for linear optimization with high adaptability to quantum computers", Mohammadhossein Mohammadisiahroudi, Ramin Fakhimi, Zeguan Wu*, and Tamás Terlaky.
SIAM Journal on Optimization.

4. "Optimal Scaling Quantum Interior Point Method for Linear Optimization", Mohammadhossein Mohammadisiahroudi, Zeguan Wu*, Pouya Sampourmahani, Jun-Kai You, and Tamás Terlaky.
IEEE Quantum Week 2025.

5. "Improvements to quantum interior point method for linear optimization", Mohammadhossein Mohammadisiahroudi, Zeguan Wu*, Brandon Augustino, Arielle Carr, and Tamás Terlaky.
ACM Transactions on Quantum Computing.

6. "An inexact feasible quantum interior point method for linearly constrained quadratic optimization", Zeguan Wu*, Mohammadhossein Mohammadisiahroudi, Brandon Augustino, Xiu Yang, and Tamás Terlaky.
Entropy.

7. "Quantum-enhanced regression analysis using state-of-the-art QLSAs and QIPMs", Mohammadhossein Mohammadisiahroudi, Zeguan Wu*, Brandon Augustino, Arielle Carr and Tamás Terlaky.
ACM/IEEE Quantum Workshop.

8. "Network design for silent link user equilibrium", Xu Chen, Zeguan Wu*, and Xuan Di.
Research Reports on Computer Science.

Conference

1. "Quantum interior points methods for linear optimization", SIAM-NNP Annual 2024, Rochester, NY.

2. "A quantum dual logarithmic barrier method for linear optimization", MOPTA 2024, Bethlehem, PA.

3. "A quantum dual logarithmic barrier method for linear optimization", SIAM Annual 2024, Spokane, WA.

4. "A detailed implementation of QLSA for linear system problem in tensor format", INFORMS Optimization Society Conference 2024, Houston, TX.

5. "An inexact feasible quantum interior point method for linear and quadratic optimization", INFORMS Annual 2023, Phoenix, AZ.

6. "An inexact feasible quantum interior point method for linear and quadratic optimization", MOPTA 2023, Bethlehem, PA.

7. "An inexact feasible quantum interior point method for linearly constrained quadratic optimization", IISE Annual 2023, New Orleans, LA.

8. "An inexact feasible quantum interior point method for linearly constrained quadratic optimization", Mid-Atlantic NA-Day 2022, Philadelphia, PA.

9. "A preconditioned quantum interior point method for linear optimization", Flash Talk, INFORMS Annual 2022, Indianapolis, IN.

10. "A preconditioned quantum interior point method for linear optimization", Poster, ICCOPT & MOPTA 2022, Bethlehem, PA.

Teaching Experience

Lehigh University

- 2023S ISE 111 Engineering Probability, TA.
- 2022F ISE 406 Introduction to Mathematical Optimization, TA.
- 2022S ISE 305/404 Simulation, TA.
- 2021F ISE 365/465 Applied Data Mining, TA.

Others

- 2023 Gene Golub SIAM Summer School, Optimization Laboratory Tutorial.

Academic Service

- Session Chair, MOPTA, 2024.
- Session Chair, SIAM Annual, 2024.
- Session Chair, INFORMS Optimization Society Conference, 2024.
- Session Chair, SIAM-NNP Annual, 2023, 2024.
- Session Chair, INFORMS Annual, 2023, 2024.
- Staff, Gene Golub SIAM Summer School, 2023.
- Session Chair, IISE Annual, 2023.
- Vice President, Lehigh University INFORMS Student Chapter, 2022–2023.
- Volunteer, ICCOPT Conference, 2022.
- Reviewer for European Journal of Operational Research.
- Reviewer for IEEE Transactions on Quantum Engineering.
- Reviewer for Journal of Computational Science.
- Reviewer for Optimization and Engineering.
- Reviewer for Scientific Reports.
- Reviewer for IEEE Quantum Week workshop.

Award

- Los Alamos National Laboratory Quantum Computing Summer School Fellowship (2023).
- Rossin Professional Development Program (2023).

References

- Tamás Terlaky, Professor of Industrial and Systems Engineering, Lehigh University.
- Xiu Yang, Associate Professor of Industrial and Systems Engineering, Lehigh University.
- Marc Vuffray, Staff Research Scientist, Los Alamos National Laboratory.
- Junyu Liu, Assistant Professor of Computer Science, University of Pittsburgh.